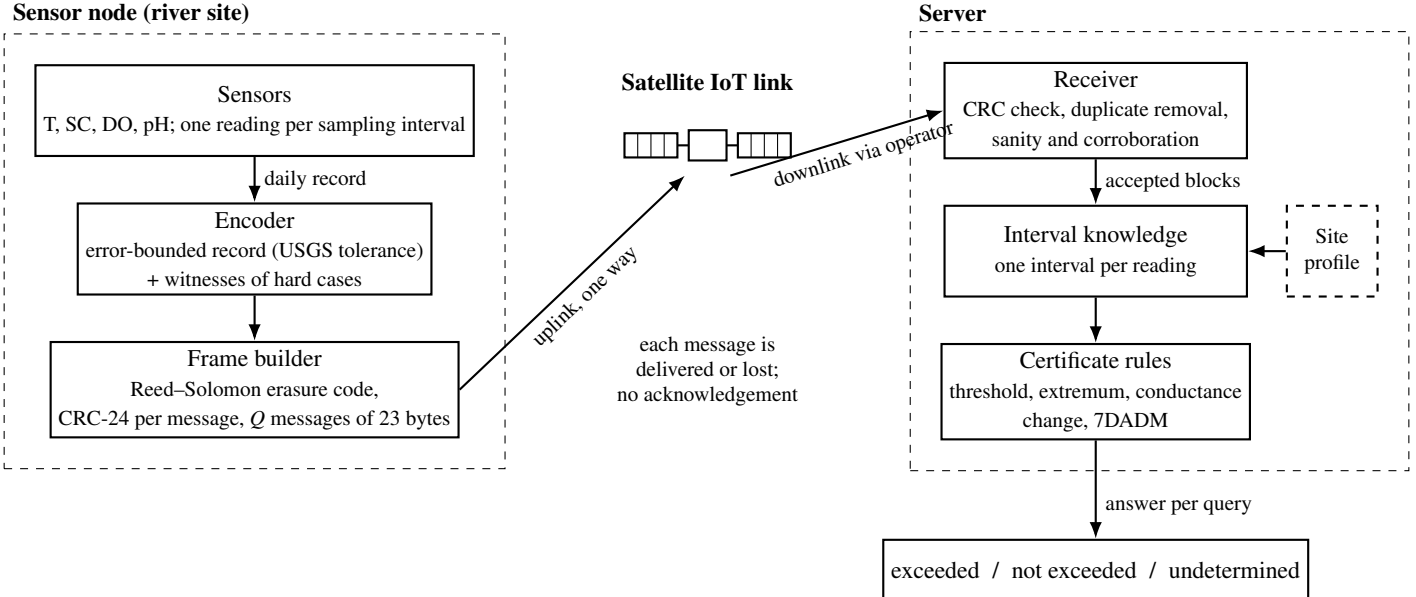
Sensor node (river site)
Sensors
T, SC, DO, pH; one reading per sampling interval
daily record
Encoder
error-bounded record (USGS tolerance)
+ witnesses of hard cases
Frame builder
Reed–Solomon erasure code,
CRC-24 per message, Q messages of 23 bytes
uplink, one way
Satellite IoT link
each message is
delivered or lost;
no acknowledgement
downlink via operator
Server
Receiver
CRC check, duplicate removal,
sanity and corroboration
accepted blocks
Interval knowledge
one interval per reading
Site
profile
Certificate rules
threshold, extremum, conductance
change, 7DADM
answer per query
exceeded / not exceeded / undetermined